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Income, Education and Body Mass Index in England: Evidence from the Health Survey for England 2021

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ABSTRACT

Socioeconomic inequalities are known to influence obesity, however the relative contributions of income, education and age to body mass index (BMI) are still unclear in England. This cross-sectional study examined the relationship between equivalised household income and BMI among adults using data from the Health Survey for England 2021 (n = 4,292). The analysis aimed to assess how income relates to BMI while accounting for age, sex and educational attainment. Descriptive statistics showed that BMI was higher among lower-income groups, with the lowest income quintile having the highest mean BMI. Non-parametric tests confirmed significant differences in BMI across income and education groups and Spearman's correlation showed a weak positive association between age and BMI. Multiple linear regression showed that, after adjusting for age, sex and education, income was not a statistically significant predictor of BMI. Higher educational attainment and older age were associated with BMI variation. These findings suggest that education and age are stronger determinants of BMI than household income alone, highlighting the complex interplay of socioeconomic and demographic factors in this study. The results have implications for public health interventions aimed at reducing obesity and associated health inequalities, emphasising the importance of addressing educational disparities alongside income-based initiatives.

1 INTRODUCTION

1.1 BACKGROUND

Body mass index (BMI), a measure of body fat based on weight relative to height, is a key indicator of obesity, which increases the risk of cardiovascular disease, type 2 diabetes, certain cancers and reduced quality of life (Ahmed & Mohammed, 2025). Socioeconomic factors, including income and education, influence BMI through multiple pathways, such as access to nutritious foods, opportunities for physical activity and engagement with health-promoting behaviours (Lee et al., 2019). BMI also varies with age and sex due to differences in body composition and lifestyle (He et al., 2018). Understanding how these factors interact is essential for identifying populations at risk and informing public health interventions.

1.2 RATIONALE FOR THE STUDY

Despite extensive research on obesity, there are still gaps in understanding of how income and education jointly influence BMI in a nationally representative UK adult population. Previous studies often focus on obesity prevalence rather than continuous BMI values or examine socioeconomic factors individually rather than simultaneously. The Health Survey for England (HSE) 2021 provides detailed, up-to-date data on demographic, socioeconomic and health measures for adults across England, making it an ideal dataset to explore these relationships. This study aims to fill these gaps by examining BMI continuously across income quintiles while controlling for education, age and sex, offering insights into socioeconomic inequalities in body weight.

1.3 Aims and Objectives

The aim of this study is to investigate the association between household income and BMI among adults in England, accounting for the influence of age, sex and educational attainment. To achieve this aim, the study will:

1. Describe the distribution of BMI across income quintiles.
2. Assess whether significant differences in BMI exist between income groups using appropriate statistical analyses.
3. Examine the role of age, sex and education as covariates influencing the income-BMI relationship.

1.4 Hypotheses

H1. BMI differs significantly across equivalised income quintiles.

H2. Individuals in lower income quintiles have higher BMI than those in higher quintiles.

H3. The association between income and BMI is influenced by sex, age and educational attainment.

2 LITERATURE REVIEW

2.1 SOCIOECONOMIC STATUS AND OBESITY

There is strong evidence that socioeconomic status (SES) is inversely associated with obesity and BMI. In the UK, adults living in the most deprived fifth of neighbourhoods had an obesity prevalence of ~35.9%, compared with ~20.5% in the least deprived fifth (Office for Health Improvement and Disparities, 2024). A review of UK studies found multiple SES indicators, including educational attainment, occupational class and area-level deprivation, were linked to higher obesity risk (El-Sayed et al., 2012). Recent studies confirm a social gradient in excess weight, noting that "the most socio-economically disadvantaged groups are affected the most by excess weight and obesity," with heterogeneity by region and gender (Candio et al., 2023). However, few studies analyse BMI as a continuous outcome across income quintiles, leaving gaps for further analyses.

2.2 MECHANISMS LINKING SOCIOECONOMIC STATUS TO BMI

Several mechanisms may explain the link between low SES and higher BMI. Food access and affordability are important as households with lower incomes often face higher costs for healthy foods and may rely on energy-dense, nutrient-poor options. In England, adults in deprived areas are almost twice as likely to live with obesity compared with those in less deprived areas (Adams, 2020). Physical environment and opportunities for activity are also relevant. Lower-income neighbourhoods may lack safe spaces for exercise or green space, reducing physical activity (Cronin et al., 2022). Psychological stress and health literacy further contribute, as chronic stress may influence eating behaviours and fat distribution, and lower education may limit engagement with health-promoting behaviours (Kumar et al., 2022; Shahid et al., 2022). These mechanisms provide an explanation for the observed association between income and BMI.

2.3 Age, Sex and Education as Covariates

Age, sex and education are important covariates that may influence or partially explain the relationship between income and BMI. BMI tends to increase with age before plateauing or declining in older adulthood. In the UK, overweight/obesity peaks at 55-64 years (~72.8%) and declines in the 65-74 age group (Office for Health Improvement and Disparities, 2024). Sex differences are also relevant, with some evidence that SES-obesity gradients are stronger in women than men (Autret & Bekelman, 2024). Educational attainment acts both as a component of SES and as a potential moderator through health literacy, lifestyle choices and career trajectories. Individuals with no school qualifications have significantly higher odds of higher BMI compared with degree holders (Booth et al., 2017). Including these variables is therefore important for unbiased assessment of income-BMI associations.

2.4 Gaps in the Literature

Despite strong evidence, gaps remain. Many studies focus on obesity categories rather than continuous BMI and fewer examine mean or median BMI across income quintiles. Analyses often study income or education separately, with fewer considering them alongside age and sex in nationally representative UK samples. Non-parametric analyses or methods adjusting for non-normal BMI are also underreported. Much evidence comes from older datasets, so up-to-date analyses using HSE 2021 are warranted. Cross-sectional designs predominate, limiting causal inference.

2.5 Summary

In summary, lower SES, indicated by lower income, education or area deprivation, is consistently linked with higher BMI. Mechanisms include poor diet, reduced physical and psychosocial stress. Age, sex and education are important covariates, yet gaps remain regarding continuous BMI, covariate modelling and contemporary data. This review justifies examining the relationship between income and BMI while controlling for these factors for adults in England.

3 METHODOLOGY

3.1 DATA SOURCE

This study used data from the HSE 2021, an annual cross-sectional survey conducted by NatCen Social Research (NHS, 2024). The survey used a multistage stratified probability sampling design to achieve proportional representation by region, age and income. Data were gathered through household interviews and nurse visits, with height and weight measured objectively. Ethical approval for HSE 2021 was granted and all participants gave informed consent. The original dataset contained 7120 individuals.

3.2 VARIABLES

The dependent variable was BMI. The main independent variable was equivalised household income quintile, adjusted for household size and composition. Covariates were sex, age group and highest educational qualification. The variables used are summarised in Table 1.

Table 1

Study Variables and Coding

Variable Type	Variable Name (SPSS)	Description	Coding / Categories
Dependent	bmisr	Body Mass Index (BMI), calculated as weight (kg) divided by height squared (m^2)	Continuous
Independent	eqv5	Equivalised household income quintile, adjusted for household composition	1 = Lowest income, 2 = Second lowest, 3 = Middle, 4 = Second highest, 5 = Highest
Covariate	sex	Biological sex of respondent	1 = Male, 2 = Female
	age16g5	Age group (five-year bands, from age 16 onwards)	17 categories ranging from 16-17 to 90+
	topqual4	Highest educational qualification attained	1 = NVQ4/NVQ5/Degree, 2 = Below degree level 3 = No qualification

3.3 Data Preparation

Data were screened and cleaned using IBM SPSS Statistics (v29). Adults under 16 years were excluded, leaving 5880 participants. Cases with missing data for any of the variables or implausible BMI values (<10 or >125 kg/m^2) were removed using listwise deletion, resulting in a final analytic sample of 4292 adults. Categorical variables were checked for consistency and labelled appropriately. Extreme values within plausible ranges were retained, reflecting genuine population variation (Frost, 2019).

3.4 STATISTICAL ANALYSIS

Analyses were performed in SPSS. Statistical significance was defined as $p < 0.05$.

3.4.1 Descriptive Analysis

Descriptive statistics were used to summarise the demographic composition of the sample and the distribution of BMI. For continuous variables, the mean, median, standard deviation, minimum and maximum were reported. Categorical variables were summarised using frequencies and percentages. Boxplots were generated to visualise BMI distribution across income quintiles.

3.4.2 Assumption Testing

Normality was assessed and illustrated using the Kolmogorov-Smirnov test and a histogram respectively, while Levene's test examined homogeneity of variance. Significant deviations from normality and unequal variances justified the use of non-parametric methods for group comparisons.

3.4.3 Inferential Analysis

A Kruskal-Wallis H test assessed differences in BMI across income quintiles. Where significant differences were found, post-hoc Mann-Whitney U tests identified which quintiles differed significantly.

3.4.4 Covariate and Exploratory Analysis

Demographic effects on BMI were analysed using non-parametric tests: Spearman's correlation for age, Mann-Whitney U for sex and Kruskal-Wallis for education. An exploratory multiple linear regression was also run to assess the combined effects of income, age, sex and education on BMI, despite departures from normality and equal variances.

4 RESULTS

4.1 DESCRIPTIVE STATISTICS

4.1.1 Sample Characteristics

The final sample consisted of slightly more female participants (52.7%) than male participants (47.3%). Age distribution was broad, with the largest groups aged 55-59 (11.1%) and 70-74 (9.6%). Educational attainment varied: 36.8% held a degree or higher qualification, 50.7% had below degree-level education and 12.5% reported no formal qualifications. Sample characteristics are summarised in Table 2.

Table 2

Sample Characteristics of Analytic Sample

Variable	Category	n	%
Sex	Male	2030	47.3
	Female	2262	52.7
Age	16-24	230	5.4
	25-44	1191	27.7
	45-64	1624	37.8
	65-74	795	18.5
	75+	542	12.6
Education	NVQ4/NVQ5/Degree	1581	36.8
	Below degree	2174	50.7
	No qualification	537	12.5

Note. Percentages may not sum to 100 due to rounding. Age groups have been collapsed for clarity.

4.1.2 BMI Distribution by Income Quintile

The overall mean BMI was 26.60, with a median of 25.70. Descriptive statistics by income quintile are presented in Table 3. Participants in the lowest quintile had the highest mean BMI (27.02), whereas those in the highest quintile had the lowest (26.02), suggesting a modest inverse relationship between income and BMI, illustrated in Figure 1.

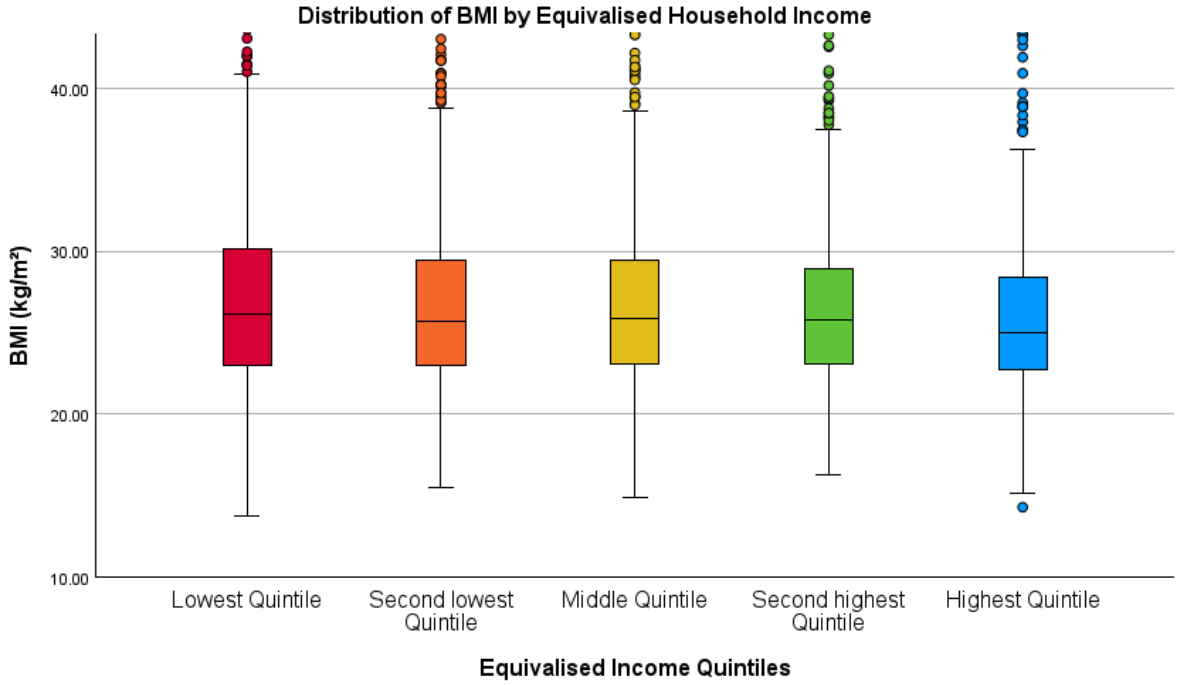
Table 3

BMI Descriptive Statistics by Equivalised Income Quintile

Income Quintile	n	Mean	SD	Median	Min	Max
Lowest	701	27.02	5.64	26.15	13.73	51.00
Second lowest	994	26.71	5.72	25.71	15.52	73.81
Middle	860	26.89	5.70	25.89	14.92	61.04
Second highest	919	26.49	5.11	25.76	16.29	68.09
Highest	818	26.02	5.93	24.97	14.29	125.00

Figure 1

Boxplot of BMI by Equivalised Income Quintile



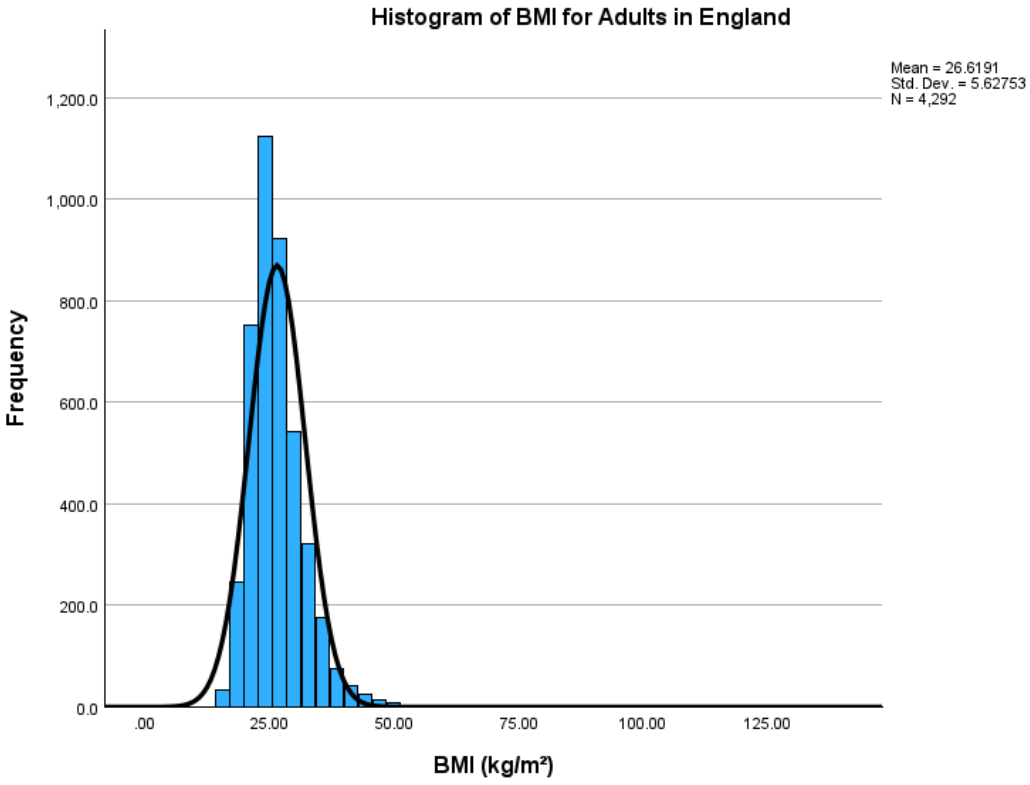
Note. The boxplot shows the median, interquartile range and outliers for each income quintile. Quintiles are colour-coded from lowest to highest (red orange, yellow, green, blue). The y-axis is truncated to 10-40 kg/m^2 for readability; some outliers beyond this range are not shown.

4.1.3 Normality and Variance Checks

The Kolmogorov-Smirnov test showed a significant deviation from normality ($D = 0.083$, $p < 0.001$), confirming that BMI was not normally distributed. As shown in Figure 2, the distribution was positively skewed, with several extreme values contributing to the asymmetry. Levene's test also indicated heterogeneity of variances across income quintiles ($F = 4.407$, $p = 0.001$), supporting the use of non-parametric methods for inferential analyses.

Figure 2

Histogram Showing the Positively Skewed Distribution of BMI



Note. The histogram illustrates the distribution of BMI across the analytic sample (n = 4292). The distribution is positively skewed, with several extreme values contributing to asymmetry.

4.2 INFERENTIAL STATISTICS

4.2.1 Kruskal-Wallis Test for Income

The Kruskal-Wallis H test revealed a statistically significant difference in BMI across income quintiles, H(4) = 18.106, p = 0.001. As shown in Table 4, participants in lower-income quintiles had higher mean BMI ranks than those in higher-income quintiles.

Table 4

Mean ranks of BMI by Income Quintile

Equivalised income quintile	Mean Rank
Lowest	2241.36
Second Lowest	2162.84
Middle	2200.47
Second Highest	2140.52
Highest	1995.34

Note. Results are from a Kruskal-Wallis H test examining differences in BMI across income quintiles (n = 4292).

4.2.2 Post-hoc Analysis

Pairwise Mann-Whitney U tests identified significant differences mainly between the highest income quintile and all lower quintiles. Participants in the lowest, second-lowest, middle and second-highest groups all had significantly higher BMI than those in the highest quintile (p-values < 0.001-0.012). These results demonstrate a clear negative gradient between income and BMI.

4.2.3 Covariate Analysis

4.2.3.1 Age

The Spearman's rank-order correlation indicated a weak but statistically significant positive association between BMI and age (ρ = 0.092, p < 0.001), suggesting that BMI tends to increase slightly as age increases.

4.2.3.2 Sex

Differences in BMI between males and females were assessed using the Mann-Whitney U test. The analysis showed a statistically significant difference (U = 2,120,226.5, Z = -4.335, p < 0.001), with males (mean rank = 2,233.05) having higher BMI than females (mean rank = 2,068.82), as summarised in Table 5.

Table 5

Mann-Whitney U Test Results for BMI by Sex

Sex	n	Mean Rank	Sum of Ranks
Male	2,535	2845.48	7,213,280.00
Female	2,892	2598.75	7,515,598.00
Total	5,427		

Note. Mann-Whitney U = 2,120,226.5, Z = -4.335, p < 0.001.

4.2.3.3 Educational Attainment

A Kruskal-Wallis test examined BMI differences across educational attainment levels. Results were statistically significant (H(2) = 54.350, p < 0.001), with participants holding no qualifications showing the highest BMI and those with higher education having the lowest. This is shown in Table 6.

Table 6

Kruskal-Wallis Test Results for BMI by Educational Attainment

Education Level	n	Mean Rank
NVQ4/NVQ5/Degree or equivalent	1,581	2,001.03
Below degree	2,174	2,219.86
No qualification	537	2,277.81

Note. Kruskal-Wallis H(2) = 35.438, p < 0.001.

These findings indicate that BMI varies significantly with age, sex and education. BMI tends to rise modestly with age, is generally higher among males and decreases as education level increases. These demographic influences are therefore important when interpreting income-related BMI patterns.

4.2.4 Multiple Linear Regression

The multiple linear regression model was statistically significant, $F(4, 4287) = 9.099$, $p < 0.001$, but explained only a small proportion of BMI variance ($R^2 = 0.008$, Adjusted $R^2 = 0.007$). Regression coefficients are shown in Table 7.

Table 7

Multiple Linear Regression Predicting BMI

Predictor	B	Std. Error	Beta	t	p
Constant	25.919	0.523	-	49.590	<0.001
Equivalised Income Quintile	-0.128	0.068	-0.031	-1.887	0.059
Educational Attainment (3 groups)	0.324	0.142	0.038	2.276	0.023
Sex	-0.202	0.172	-0.018	-1.176	0.239
Age (16+ 5-year bands)	0.089	0.025	0.057	3.603	<0.001

Note. Dependent variable: BMI. $R^2 = 0.008$, Adjusted $R^2 = 0.007$.

Age and education were significant predictors: higher age was associated with slightly higher BMI and higher educational attainment with slightly lower BMI. Income and sex were not significant when adjusting for other variables. Overall, while BMI differences exist across income groups, age and education appear to be stronger determinants.

5 DISCUSSION

5.1 INCOME AND BMI

A modest inverse relationship between income and BMI was observed. Lower-income quintiles had higher BMI, with post-hoc tests showing the greatest differences between the highest and all lower quintiles, demonstrating a clear socioeconomic gradient. These findings are consistent with prior research showing that socioeconomic disadvantage is associated with excess body weight, reflecting differential access to nutritious foods, opportunities for physical activity and other health-promoting resources. However, after adjusting for age, sex and education, income was no longer a significant predictor, suggesting demographic factors, particularly education and age, explain much of the variation. Income may influence BMI indirectly but is less robust than other determinants.

5.2 EDUCATION AND BMI

Education emerged as a significant independent predictor of BMI, with higher qualifications associated with lower BMI. Participants with no formal qualifications showed the highest BMI, highlighting the role of education in shaping health outcomes. Education may influence BMI through multiple pathways, including better health literacy, greater knowledge of nutrition, healthier lifestyle behaviours and career trajectories that facilitate access to resources promoting physical health (Booth et al., 2017; Kumar et al., 2022). These findings show the importance of addressing educational disparities alongside income-based interventions in public health strategies.

5.3 AGE AND SEX

Age showed a weak but significant positive association with BMI, reflecting gradual weight gain across adulthood due to metabolic changes and lifestyle factors (Office for Health Improvement and Disparities, 2024). Sex differences were observed, with males exhibiting slightly higher BMI than females. However, sex was not a significant predictor after adjusting for other covariates. This suggests that the influence of sex is smaller than that of education or age and may be context-dependent, potentially interacting with other socioeconomic factors.

5.4 Methodological Considerations

Several methodological factors should be considered when interpreting these findings. BMI was positively skewed with extreme values, and variance was heterogeneous across income quintiles, justifying non-parametric analyses. Consequently, the multiple linear regression results should be interpreted cautiously due to violations of normality and homoscedasticity, which may affect its reliability (Qu, 2024). The cross-sectional design precludes causal inference, and the regression model explained only a small proportion of variance in BMI, suggesting unmeasured factors, such as diet, physical activity, smoking and chronic conditions, contribute substantially (Savitz & Wellenius, 2022; Gao, 2023). Additionally, self-reported income and education may introduce reporting bias, and the exclusion of participants with missing data could create selection bias, although the final sample remained broadly representative (van der Steen et al., 2019; Millard et al., 2022).

5.5 Public Health Implications

These results show the significance of socioeconomic and educational factors in shaping BMI and health inequalities. Interventions aiming to reduce obesity should not focus solely on income but also target education and health literacy. Approaches could include community-based nutrition education, promotion of physical activity in deprived areas and tailored interventions for groups with lower educational qualifications. By addressing both educational and income-related disparities, interventions may more effectively mitigate the burden of obesity in England.

5.6 Future Research

Future studies should adopt longitudinal designs to clarify causal relationships between socioeconomic factors and BMI. Additional covariates, like diet, physical activity, mental health and environmental factors, should be incorporated to better explain variation in BMI. Analyses stratified by region, ethnicity or gender may further elucidate population subgroups at heightened risk of excess weight, enabling more targeted interventions.

6 CONCLUSION

6.1 KEY FINDINGS

Using HSE 2021 data, this study examined associations between household income, educational attainment, age, sex and BMI in adults. Lower income was modestly linked to higher BMI descriptively, but this association weakened after adjusting for age, sex and education. Education was an independent predictor, with higher qualifications associated with lower BMI. Age showed a weak positive association, while sex differences were minimal. Overall, age and education explained more variation in BMI than income or sex.

6.2 LIMITATIONS

The cross-sectional design prevents causal inference and the regression model explained only a small proportion of variance in BMI, suggesting that unmeasured factors like diet, physical activity, mental health and environmental influences are relevant. BMI was positively skewed, and self-reported income and education may introduce bias. Excluding participants with missing data may have caused selection effects.

6.3 IMPLICATIONS AND FUTURE RESEARCH

The findings show the effect of socioeconomic and educational inequalities in BMI and public health. Interventions should improve health literacy, promote healthy behaviours and address educational disparities alongside income-based strategies. Targeted programmes for low-education groups and older adults may be particularly effective in mitigating obesity risk. Future research should use longitudinal designs, integrate lifestyle and environmental variables and examine differences across gender, region and ethnicity to identify high-risk subgroups and guide tailored public health interventions.

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8 APPENDIX

8.1 DESCRIPTIVE STATISTICS

Descriptives

			Statistic	Std. Error
(D) Self-reported BMI	Mean		26.6191	.08590
	95% Confidence Interval for Mean	Lower Bound	26.4507	
		Upper Bound	26.7875	
	5% Trimmed Mean		26.2433	
	Median		25.7031	
	Variance		31.669	
	Std. Deviation		5.62753	
	Minimum		13.73	
	Maximum		125.00	
	Range		111.27	
	Interquartile Range		6.25	
	Skewness		2.529	.037
	Kurtosis		25.681	.075

Sex

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	2030	47.3	47.3	47.3
	Female	2262	52.7	52.7	100.0
	Total	4292	100.0	100.0	

(D) Age 16+, 5 year bands

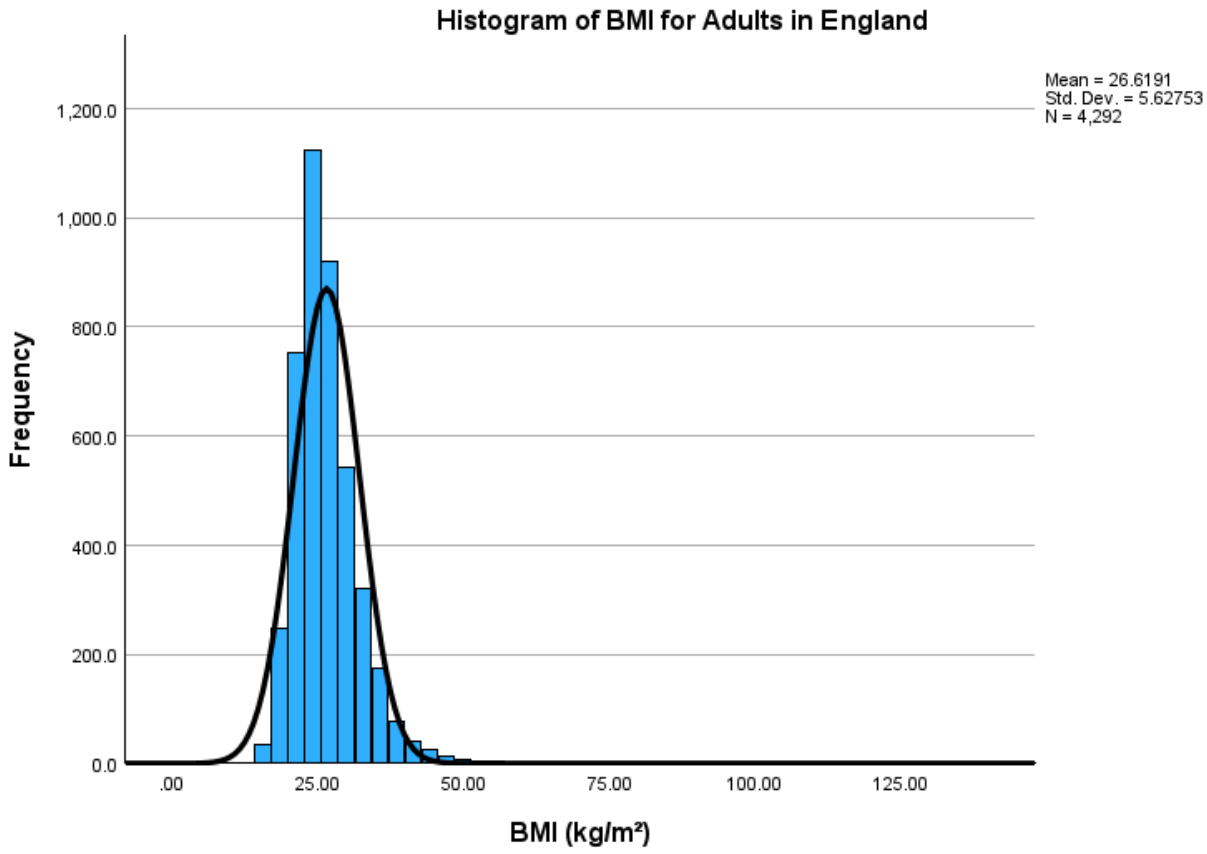
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	16-17 year olds	67	1.6	1.6	1.6
	18-19 year olds	55	1.3	1.3	2.8
	20-24 year olds	108	2.5	2.5	5.4
	25-29 year olds	214	5.0	5.0	10.3
	30-34 year olds	279	6.5	6.5	16.8
	35-39 year olds	360	8.4	8.4	25.2
	40-44 year olds	338	7.9	7.9	33.1
	45-49 year olds	310	7.2	7.2	40.3
	50-54 year olds	386	9.0	9.0	49.3
	55-59 year olds	478	11.1	11.1	60.5
	60-64 year olds	360	8.4	8.4	68.8
	65-69 year olds	383	8.9	8.9	77.8
	70-74 year olds	412	9.6	9.6	87.4
	75-79 year olds	302	7.0	7.0	94.4
	80-84 year olds	155	3.6	3.6	98.0
	85-89 year olds	67	1.6	1.6	99.6
	90+	18	.4	.4	100.0
	Total	4292	100.0	100.0	

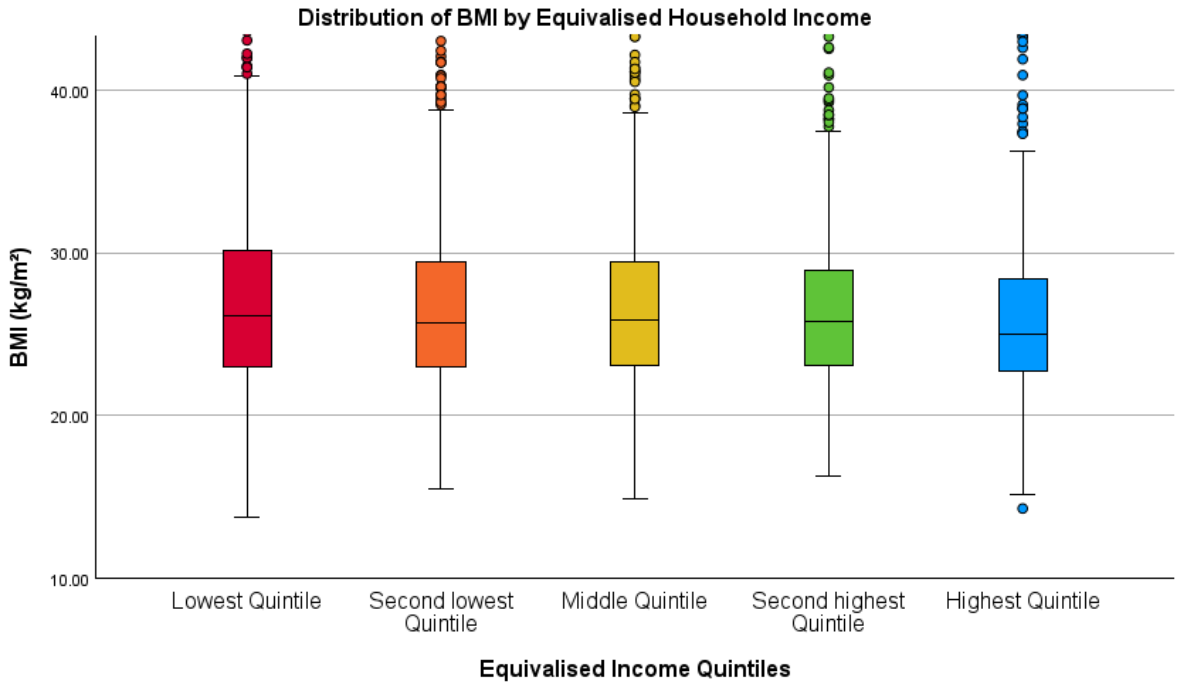
(D) Highest Educational Qualification, 3 groups

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	NVQ4/NVQ5/Degree or equiv	1581	36.8	36.8	36.8
	Below degree	2174	50.7	50.7	87.5
	No qualification	537	12.5	12.5	100.0
	Total	4292	100.0	100.0	

(D) Equivalised Income Quintiles

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Lowest Quintile	701	16.3	16.3	16.3
	Second lowest Quintile	994	23.2	23.2	39.5
	Middle Quintile	860	20.0	20.0	59.5
	Second highest Quintile	919	21.4	21.4	80.9
	Highest Quintile	818	19.1	19.1	100.0
	Total	4292	100.0	100.0	





8.2 Assumption Testing

Tests of Normality

	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
(D) Self-reported BMI	.083	4292	<.001	.880	4292	<.001

a. Lilliefors Significance Correction

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
(D) Self-reported BMI	Based on Mean	4.407	4	4287	.001
	Based on Median	4.080	4	4287	.003
	Based on Median and with adjusted df	4.080	4	4108.913	.003
	Based on trimmed mean	4.315	4	4287	.002

8.3 Inferential Analysis

Kruskal-Wallis Test

Ranks

	(D) Equivalised Income Quintiles	N	Mean Rank
(D) Self-reported BMI	Lowest Quintile	701	2241.36
	Second lowest Quintile	994	2162.84
	Middle Quintile	860	2200.47
	Second highest Quintile	919	2140.52
	Highest Quintile	818	1995.34
	Total	4292	

Test Statistics[a,b]

	(D) Self-reported BMI
Kruskal-Wallis H	18.106
df	4
Asymp. Sig.	.001

a. Kruskal Wallis Test

b. Grouping Variable: (D) Equivalised Income Quintiles

Mann-Whitney Test

Ranks

	(D) Equivalised Income Quintiles	N	Mean Rank	Sum of Ranks
(D) Self-reported BMI	Lowest Quintile	701	805.76	564840.50
	Highest Quintile	818	720.78	589599.50
	Total	1519		

Test Statistics[a]

	(D) Self-reported BMI
Mann-Whitney U	254628.500
Wilcoxon W	589599.500
Z	-3.764
Asymp. Sig. (2-tailed)	<.001

a. Grouping Variable: (D) Equivalised Income Quintiles

Ranks

	(D) Equivalised Income Quintiles	N	Mean Rank	Sum of Ranks
(D) Self-reported BMI	Second lowest Quintile	994	938.21	932581.00
	Highest Quintile	818	867.97	709997.00
	Total	1812		

Test Statistics[a]

	(D) Self-reported BMI
Mann-Whitney U	375026.000
Wilcoxon W	709997.000
Z	-2.844
Asymp. Sig. (2-tailed)	.004

a. Grouping Variable: (D) Equivalised Income Quintiles

Ranks

	(D) Equivalised Income Quintiles	N	Mean Rank	Sum of Ranks
(D) Self-reported BMI	Middle Quintile	860	878.76	755736.50
	Highest Quintile	818	798.22	652944.50
	Total	1678		

Test Statistics[a]

	(D) Self-reported BMI
Mann-Whitney U	317973.500
Wilcoxon W	652944.500
Z	-3.404
Asymp. Sig. (2-tailed)	<.001

a. Grouping Variable: (D) Equivalised Income Quintiles

Ranks

	(D) Equivalised Income Quintiles	N	Mean Rank	Sum of Ranks
(D) Self-reported BMI	Second highest Quintile	919	897.60	824891.50
	Highest Quintile	818	836.87	684561.50
	Total	1737		

Test Statistics[a]

	(D) Self-reported BMI
Mann-Whitney U	349590.500
Wilcoxon W	684561.500
Z	-2.519
Asymp. Sig. (2-tailed)	.012

a. Grouping Variable: (D) Equivalised Income Quintiles

Correlations

			(D) Self-reported BMI	(D) Age 16+, 5 year bands
Spearman's rho	(D) Self-reported BMI	Correlation Coefficient	1.000	.092**
		Sig. (2-tailed)	.	<.001
		N	4292	4292
	(D) Age 16+, 5 year bands	Correlation Coefficient	.092**	1.000
		Sig. (2-tailed)	<.001	.
		N	4292	4292

**. Correlation is significant at the 0.01 level (2-tailed).

Test Statistics[a]

	(D) Self-reported BMI
Mann-Whitney U	2120226.500
Wilcoxon W	4679679.500
Z	-4.335
Asymp. Sig. (2-tailed)	<.001

a. Grouping Variable: Sex

Test Statistics[a,b]

	(D) Self-reported BMI
Kruskal-Wallis H	35.438
df	2
Asymp. Sig.	<.001

a. Kruskal Wallis Test

b. Grouping Variable: (D) Highest Educational Qualification, 3 groups

8.4 Multiple Linear Regression

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.092[a]	.008	.007	5.60641

a. Predictors: (Constant), (D) Highest Educational Qualification, 3 groups, Sex, (D) Age 16+, 5 year bands, (D) Equivalised Income Quintiles

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	25.919	.523		49.590	<.001
	(D) Equivalised Income Quintiles	-.128	.068	-.031	-1.887	.059
	(D) Age 16+, 5 year bands	.089	.025	.057	3.603	<.001
	Sex	-.202	.172	-.018	-1.176	.239
	(D) Highest Educational Qualification, 3 groups	.324	.142	.038	2.276	.023

a. Dependent Variable: (D) Self-reported BMI